



RV College of Engineering®

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Summer Internship Report

on

**“Gene Expression-Based Cancer
Classification Using Machine Learning for
Precision Oncology”**

CS376SI

Submitted by

Shreejay Pandey

1RV22CS187

*Submitted in
partial fulfillment for the award of degree
of*

BACHELOR OF ENGINEERING

in

Computer Science and Engineering,

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

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RV COLLEGE OF ENGINEERING®

(Autonomous Institution Affiliated to Visvesvaraya Technological University,
Belagavi)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Bengaluru – 560059



CERTIFICATE

Certified that the Internship work titled **“Gene Expression-Based Cancer Classification Using Machine Learning for Precision Oncology”** carried out by **Shreejay Pandey, USN: 1RV22CS187**, a bonafide student, submitted in partial fulfilment for the award of Bachelor of Engineering in **Computer Science and Engineering** of RV College of Engineering®, Bengaluru, affiliated to Visvesvaraya Technological University, Belagavi, during the year **2025–26**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The Internship report has been approved as it satisfies the academic requirement in respect of internship work prescribed for the said degree.

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This is to certify that **Shreejay Pandey, (1RV22CS187)**, a student of Computer Science and Engineering, RV College of Engineering, Bengaluru, has successfully completed an Industry Certified Internship under the Certificate Course on Healthcare Analytics conducted by the **Centre of Excellence in Computational Genomics (CoE-CG)** for 3 months duration for the academic year 2025-26.

During the internship period, the student has gained hands-on exposure to healthcare data analytics, machine learning, and deep learning techniques with applications in medical image analysis. The student demonstrated sincere effort, discipline and technical competence while carrying out the assigned tasks.

We wish him all the best in his future academic and professional endeavours.

Dr. AH Manjunatha Reddy

Professor

Head-Centre of Excellence in Computational Genomics

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EXECUTIVE SUMMARY

Cancer diagnosis and treatment planning increasingly rely on molecular-level insights to distinguish cancer types and subtypes with higher precision. With the rapid growth of high-throughput sequencing and bioinformatics, gene expression (RNA-Seq) data has become a powerful representation of tumour behaviour, capturing patterns of abnormal cellular activity through thousands of gene measurements per patient. Compared to traditional diagnosis based mainly on imaging and histopathology, gene expression analysis provides deeper biological understanding and supports the broader vision of precision oncology by enabling data-driven and personalized clinical decision-making.

Despite its importance, gene expression-based cancer classification remains challenging due to the highly complex and high-dimensional nature of genomic datasets. Each sample may contain expression values for thousands of genes, often with redundancy, noise, and variation in scale, making direct model training inefficient and prone to overfitting. This project addresses the need for an automated and reliable cancer classification system that can reduce diagnostic uncertainty, improve speed of analysis, and support early subtype identification, ultimately enabling more accurate treatment planning and better patient outcomes.

To achieve this, the project implements a complete machine learning pipeline involving preprocessing, normalization, and feature scaling, followed by dimensionality reduction using Principal Component Analysis (PCA) and autoencoder-based feature learning. Multiple supervised classifiers such as Support Vector Machine (SVM), Random Forest, Logistic Regression, and K-Nearest Neighbors (KNN) were trained and compared using evaluation metrics including accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. The results demonstrate that machine learning can effectively learn gene expression signatures for cancer classification, with feature reduction significantly improving stability and performance, and SVM achieving the best overall accuracy and generalization among the evaluated models.

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CHAPTER 1

PROFILE OF THE ORGANIZATION

1.1 ORGANIZATIONAL STRUCTURE

The project is carried out at **RV College of Engineering (RVCE), Bengaluru**, under the academic and research ecosystem of the institution, which supports advanced engineering research, innovation-driven learning, and interdisciplinary project development. RVCE is an **autonomous private engineering college affiliated to Visvesvaraya Technological University (VTU)**, offering multiple undergraduate and postgraduate programs across departments such as **Computer Science and Engineering (CSE), Information Science and Engineering (ISE), Electronics and Communication Engineering (ECE), Mechanical Engineering, and Civil Engineering**. The institutional structure is organized under the guidance of the **Principal and Dean (Academics)**, with individual departments and faculty mentors coordinating internship activities through structured review and evaluation processes, ensuring timely progress and academic quality.

The internship project titled **“Gene Expression-Based Cancer Classification Using Machine Learning for Precision Oncology”** is undertaken in a research-supported environment within RVCE, where the institution promotes applied learning in areas such as **machine learning, data analytics, and healthcare-focused computational research**. The project execution is supported through departmental supervision and access to academic infrastructure that enables students to work on real-world datasets and implement complete machine learning pipelines. This organizational setup encourages systematic development through continuous guidance, technical discussions, and milestone-based assessment.

In addition, RVCE supports advanced project development through its research culture and collaborative framework, which enables students to explore emerging domains such as **bioinformatics and precision medicine**, where machine learning techniques are increasingly used to extract meaningful patterns from high-dimensional gene expression data. The structured organizational hierarchy, combined with mentorship and research exposure, provides an effective platform for students to develop technically strong and socially relevant solutions that align with modern healthcare needs.

1.2 PRODUCTS

RVCE as an institution does not produce commercial healthcare products, but its research groups, academic laboratories, and interdisciplinary project teams regularly deliver **research outputs, software prototypes, and data-driven systems** relevant to real-world applications. In the context of this internship, the Department of Computer Science and Engineering and its associated research-driven activities support the development of applied machine learning solutions in healthcare and biomedical analytics.

The major products and deliverables associated with this project include:

- **Research prototype for cancer subtype classification** using gene expression (RNA-Seq) data, designed as a complete workflow for precision oncology support.
- **Machine learning pipeline implementation** involving preprocessing, normalization, feature scaling, and dimensionality reduction using PCA and autoencoder-based feature learning.
- **Supervised classification models and evaluation modules** including SVM, Random Forest, Logistic Regression, and KNN, with metric-based comparison using accuracy, F1-score, confusion matrix, and ROC-AUC analysis.
- **Visualization and analysis outputs** such as PCA scatter plots, ROC curves, and performance comparison summaries, supporting interpretability and result validation.
- **Documentation and reporting artifacts** including structured reports, experimental logs, and presentation-ready outputs for academic evaluation and future enhancements.

The ecosystem also encourages development of reusable software components, internal scripts, and utilities for efficient dataset handling, model training, and benchmarking, which strengthens the practical and research-oriented contribution of the work. Such project outputs align with the institution's goal of building deployable and scalable solutions in emerging domains such as **bioinformatics, precision medicine, and healthcare analytics**.

1.3 SERVICES

The combined ecosystem of RVCE and its academic research environment provides a wide range of academic and technical services that support project development and applied learning. These services are designed to strengthen student capabilities through structured education, research exposure, and access to modern tools and infrastructure. In the context of this internship, the ecosystem supports interdisciplinary projects in

domains such as **machine learning, data analytics, healthcare computing, and bioinformatics**, which directly aligns with the objective of gene expression-based cancer classification.

The key services offered include:

- **Education and training:** Undergraduate and postgraduate programmes in Computer Science and Engineering, along with exposure to core areas such as machine learning, artificial intelligence, data science, and computational biology, supported by laboratory sessions, project-based learning, and research-driven curricula.
- **Research support and academic supervision:** Faculty mentoring, technical guidance, and structured project review processes that ensure systematic progress in model development, experimentation, and validation. This includes assistance in dataset handling, algorithm selection, and performance evaluation to maintain academic quality.
- **Infrastructure and computing resources:** Access to departmental computing facilities and software environments required for implementing machine learning pipelines involving preprocessing, dimensionality reduction, training, and visualization. These resources enable repeated experimentation and benchmarking on complex biomedical datasets.
- **Capacity building and upskilling:** Workshops, seminars, certification programmes, and technical training sessions in emerging technologies such as AI/ML, healthcare analytics, and data-driven decision systems. These initiatives support continuous learning and help students gain practical exposure beyond classroom content.

The research-oriented environment specifically supports experimentation with real-world biomedical datasets and encourages development of scalable analytical workflows. This service ecosystem directly contributes to the successful implementation of the proposed cancer classification pipeline and strengthens the overall learning outcomes of the internship.

1.4 BUSINESS PARTNERS

RVCE maintains active collaborations with several academic institutions, industry partners, and research organizations that enhance the relevance and applicability of student projects. These partnerships help align academic research with real-world problem statements and emerging technological needs. Key partnerships relevant to this project include:

- **Technical University of Applied Sciences Würzburg-Schweinfurt (THWS), Germany:** A strategic academic partner involved in joint activities related to ap-

plied research, artificial intelligence, data analytics, and international academic programmes.

- **Industry-linked Centres of Excellence at RVCE:** Various CoEs collaborate with technology companies and OEMs for curriculum enrichment, internships, and applied projects, providing industry-aligned exposure and practical relevance.
- **Healthcare and data-driven research collaborations:** Academic partnerships and research initiatives that support projects in biomedical data analysis, bioinformatics, and AI-driven healthcare applications.
- **Regional industry and start-ups:** Engagement with local industries and start-ups through applied projects and internships, ensuring that research outcomes are aligned with practical requirements and societal needs.

These collaborations provide a supportive ecosystem for developing data-driven solutions and ensure that projects such as gene expression-based cancer classification remain aligned with academic rigor and real-world applicability.

1.5 FINANCIALS

RVCE, being a private autonomous institution with a long academic history, sustains its operations and growth through a combination of institutional revenue and external support. The primary sources of funding include academic fee structures, internally supported development initiatives, and sponsored research or consultancy activities carried out by various departments. In addition, industry-supported Centres of Excellence contribute significantly by enabling modern laboratories, upgraded computing infrastructure, and specialised training programmes, which indirectly strengthen research-oriented projects undertaken by students and faculty.

The Department of Biotechnology and Engineering, which provides academic support and domain expertise for this project, is strengthened through institutional research support and project-based funding mechanisms that enable laboratory development, computational resources, and interdisciplinary collaboration. Such financial frameworks indirectly support student and faculty research activities, guided projects, and shared academic infrastructure that is utilized for healthcare-oriented work such as gene expression analysis, machine learning experimentation, and precision oncology research.

1.6 MANPOWER

RVCE consists of qualified manpower at multiple levels, including academic faculty, research mentors, technical staff, and students. Faculty members play a key role by guiding

project planning, methodology selection, and evaluation processes. The technical staff contribute by supporting laboratory activities, computing access, and software environment setup required for implementation.

For this project, manpower support is provided mainly through mentoring and supervision from the Department of Computer Science and Engineering and interdisciplinary academic guidance from the Department of Biotechnology and Engineering. This collaborative structure ensures that both the computational aspects of machine learning and the biological relevance of gene expression analysis are addressed properly. Such teamwork-based manpower distribution supports smooth execution, systematic validation, and quality documentation throughout the internship.

1.7 SOCIETAL CONCERNS

RVCE promotes socially relevant research and student projects, especially those that address healthcare, sustainability, and technology for societal benefit. Cancer diagnosis and treatment planning are major healthcare concerns, and accurate classification of cancer types and subtypes plays an important role in improving clinical decision-making and treatment personalization.

This internship project directly aligns with societal needs by focusing on gene expression-based cancer classification using machine learning. By enabling automated pattern learning from transcriptomic data, the system contributes to early detection support, subtype identification, and precision oncology-based healthcare improvement. Such applications reduce diagnostic uncertainty and strengthen the overall use of modern data-driven methods for societal well-being.

1.8 PROFESSIONAL PRACTICES

RVCE encourages professional practices in academic and research work by promoting structured project execution, systematic documentation, and quality-driven evaluation. Internships and project activities are carried out using milestone-based planning, progress reviews, and reporting standards that ensure discipline and academic correctness.

In this project, professional practices include careful dataset handling, reproducible pre-processing and model training workflows, fair comparison of machine learning models, and evaluation using standard performance metrics such as accuracy, F1-score, confusion matrix, and ROC-AUC. Documentation was maintained in a formal reporting format, ensuring that the work is clearly communicated and meets institutional academic requirements. These practices strengthen both technical reliability and professionalism in research-oriented project development.

CHAPTER 2

ACTIVITIES OF THE DEPARTMENT

The project is carried out under the academic and technical environment of the Department of Computer Science and Engineering (CSE) at RV College of Engineering (RVCE), Bengaluru, with interdisciplinary guidance supporting the biomedical nature of the work. The department actively promotes teaching, research, innovation, and outreach in both core computing and advanced domains such as machine learning, data analytics, cybersecurity, and healthcare computing. The activities of the department provide a strong foundation for executing research-oriented projects that require structured experimentation, systematic evaluation, and professional documentation.

Key departmental activities that support and frame this project include:

- **Academic programmes and curriculum:** The department offers undergraduate and postgraduate programmes with strong emphasis on programming, algorithms, database systems, machine learning, and data science. Lab-intensive learning and mini-projects encourage students to apply theoretical foundations to practical problem solving, which is essential for building end-to-end machine learning workflows such as cancer classification from gene expression data.
- **Research culture and project mentorship:** Faculty members support student learning through mentoring, periodic project reviews, and guidance on methodology selection. This research culture encourages students to explore high-impact domains such as bioinformatics and precision medicine, where data-driven decision systems are increasingly important.
- **Access to tools and computational infrastructure:** The department provides the resources required for modern machine learning experimentation, including dataset handling, preprocessing pipelines, feature reduction, model training, and performance evaluation. This support enables students to test multiple models, compare results fairly, and ensure reproducibility across experiments.
- **Workshops, training programmes, and technical events:** The department frequently conducts workshops and training sessions in emerging technologies such as AI/ML, data analytics, and healthcare applications. These activities enhance

practical technical skills and expose students to industry-relevant tools and best practices.

- **Student projects and innovation environment:** Students are encouraged to undertake projects addressing real-world problems using advanced computational methods. Projects are designed to strengthen analytical thinking, software development skills, and research reporting quality. The current project on gene expression-based cancer classification reflects this objective by integrating preprocessing, dimensionality reduction, supervised learning, and evaluation for precision oncology support.

Through these activities, the department provides a strong ecosystem that supports the planning, design, implementation, validation, and documentation of technically intensive projects such as this machine learning-based cancer classification framework.

CHAPTER 3

TASKS PERFORMED

3.1 SUMMARY OF THE TASKS PERFORMED DURING THE PROJECT

The project followed a structured, phase-wise execution spanning dataset understanding, preprocessing and normalization, dimensionality reduction, supervised model training, and comprehensive evaluation—directly mapping to the objective of developing a reliable gene expression-based cancer classification framework for precision oncology. This progression transformed theoretical concepts from literature and foundational machine learning knowledge into a complete, reproducible workflow capable of handling high-dimensional RNA-Seq datasets and producing stable classification outcomes.

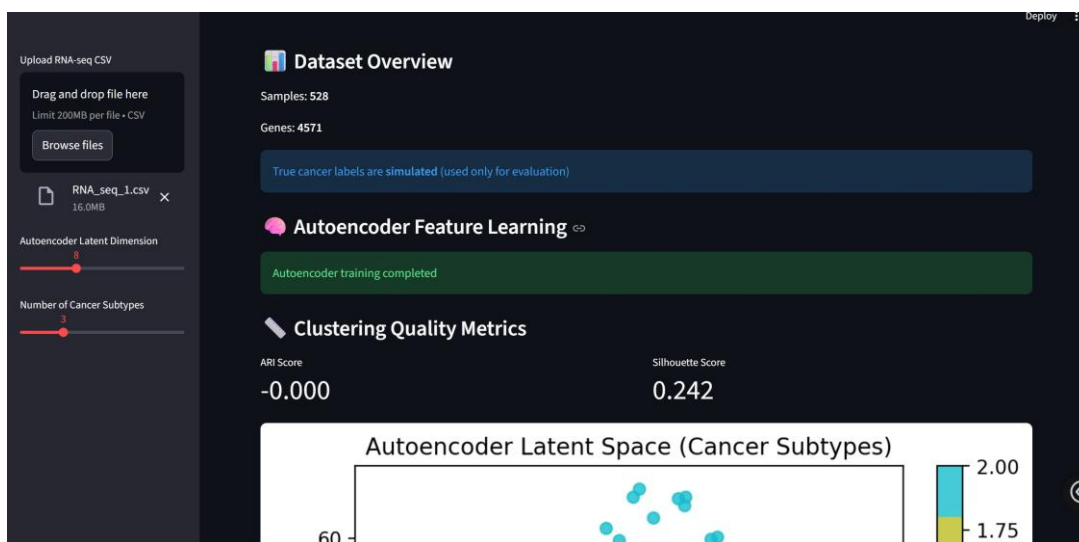


Fig. 3.1: Dataset Overview Dashboard for RNA-Seq Gene Expression Data

Each phase included iterative experimentation, parameter tuning, and validation against key performance metrics such as accuracy, F1-score, confusion matrix, and ROC-AUC to ensure robustness and generalization across cancer classes. The pipeline was designed in a modular manner so that each stage could be tested independently, improving reproducibility and making the workflow suitable for future enhancements in precision oncology.

applications.

3.1.1 Dataset Collection and Description

The dataset used in this project consists of gene expression (RNA-Seq) profiles collected from publicly available biomedical sources. Each sample corresponds to an individual patient record containing expression values of thousands of genes. These expression values represent transcriptomic signatures that capture molecular behavior and enable classification of different cancer types or subtypes. The dataset was initially analyzed to verify class distribution, the dimensionality of features, and the overall structure of the input matrix.

Table 3.1: Dataset Summary (Samples, Genes and Cancer Classes)

Parameter	Description
Dataset Type	Gene Expression (RNA-Seq)
Input Features	Thousands of gene expression values
Samples	Multiple patient samples (train + test)
Target Output	Cancer type / subtype class label
Data Source	Public biomedical repositories (TCGA/GEO)

The objective of this stage was to ensure that the dataset is correctly interpreted for supervised learning and that each sample is mapped properly to a cancer class label. This enables downstream stages such as preprocessing, dimensionality reduction, and classifier training to be applied in a consistent and reproducible manner.

3.1.2 Data Preprocessing and Normalization

Gene expression data is complex and often contains noise and scale variations across gene features. Therefore, preprocessing was performed to make the dataset suitable for machine learning algorithms. The preprocessing stage included removal of irrelevant non-numeric attributes, handling missing values if present, and transforming the dataset into a standardized numerical format. Since gene values can differ widely in magnitude, feature scaling was applied so that learning remains stable and unbiased across models.

Feature Scaling (Standardization):

$$x' = \frac{x - \mu}{\sigma}$$

where x represents the original gene expression value, μ is the mean, and σ is the standard deviation.

Table 3.2: Data Preprocessing and Feature Engineering Steps

Step	Purpose
Data Cleaning	Remove irrelevant/non-numeric fields and handle missing values
Normalization	Reduce bias due to varying gene expression scales
Feature Scaling	Improve learning stability across ML models
Train-Test Split	Ensure unbiased evaluation on unseen samples
Label Encoding	Convert cancer classes into ML-readable format

The preprocessing stage ensures that the dataset is machine learning ready and reduces technical variation that can affect training stability. It also improves fairness in model comparison since all models receive standardized feature representations.

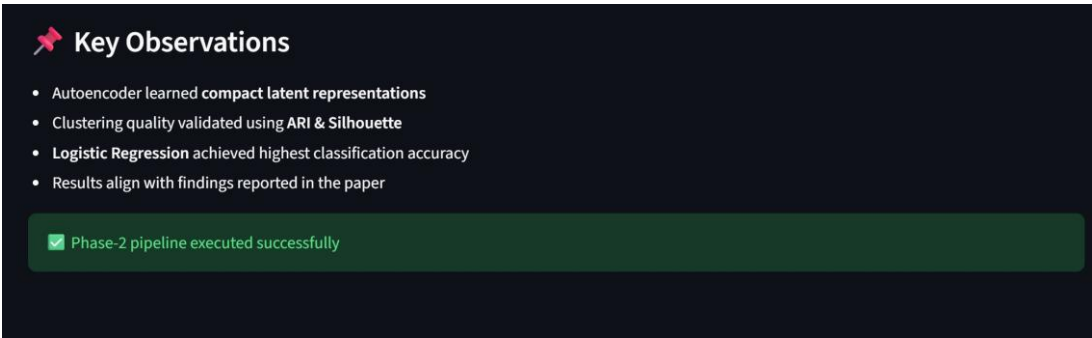


Fig. 3.2: Key Observations Dashboard and Experimental Highlights

3.1.3 Feature Reduction using PCA

Since RNA-Seq datasets contain thousands of gene features, dimensionality reduction was required to reduce redundancy and improve computational efficiency. Principal Component Analysis (PCA) was applied to transform the original high-dimensional gene space into a smaller feature space while preserving maximum variance. PCA reduces overfitting risk by compressing the dataset into orthogonal principal components, enabling efficient training and better class separability.

PCA Formulation:

$$X \in \mathbb{R}^{n \times d}$$

where n is the number of samples and d is the number of gene features.

$$C = \frac{1}{n} X^T X \quad C v = \lambda v \quad Z = X W$$

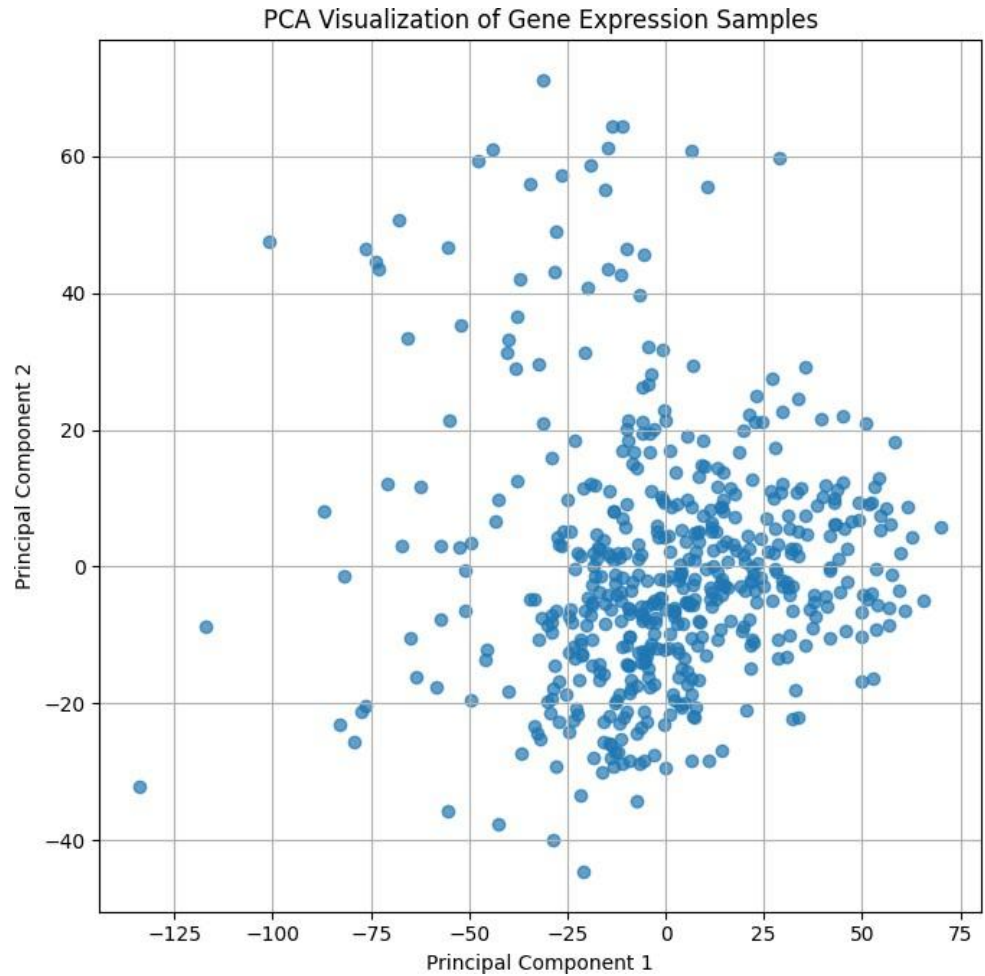


Fig. 3.3: PCA Visualization of Gene Expression Samples in Reduced Space

PCA provides visualization capability for understanding the distribution of cancer samples in reduced dimensions. This helps validate whether meaningful separation exists between cancer classes before training classifiers.

Table 3.3: PCA and Autoencoder Feature Reduction Configuration

Technique	Configuration / Role
PCA	Dimensionality reduction by preserving maximum variance
Number of Components	Selected based on variance retention and performance
Autoencoder	Learns compact nonlinear feature representation
Latent Features	Used as input for downstream classification models
Output Benefit	Reduced noise, redundancy and improved generalization

3.1.4 Autoencoder-based Feature Learning

In addition to PCA, an autoencoder-based feature learning approach was incorporated to capture nonlinear relationships among genes. Autoencoders learn compact latent representations by encoding the high-dimensional input into a reduced latent space and

reconstructing it with minimal error. This method supports better representation learning compared to linear reduction methods in cases where gene interactions are nonlinear and complex.

Autoencoder Representation:

$$h = f(Wx + b) \quad \hat{x} = g(W'h + b')$$

Reconstruction Loss (MSE):

$$L_{rec} = \frac{1}{n} \sum_{i=1}^n \|x_i - \hat{x}_i\|^2$$

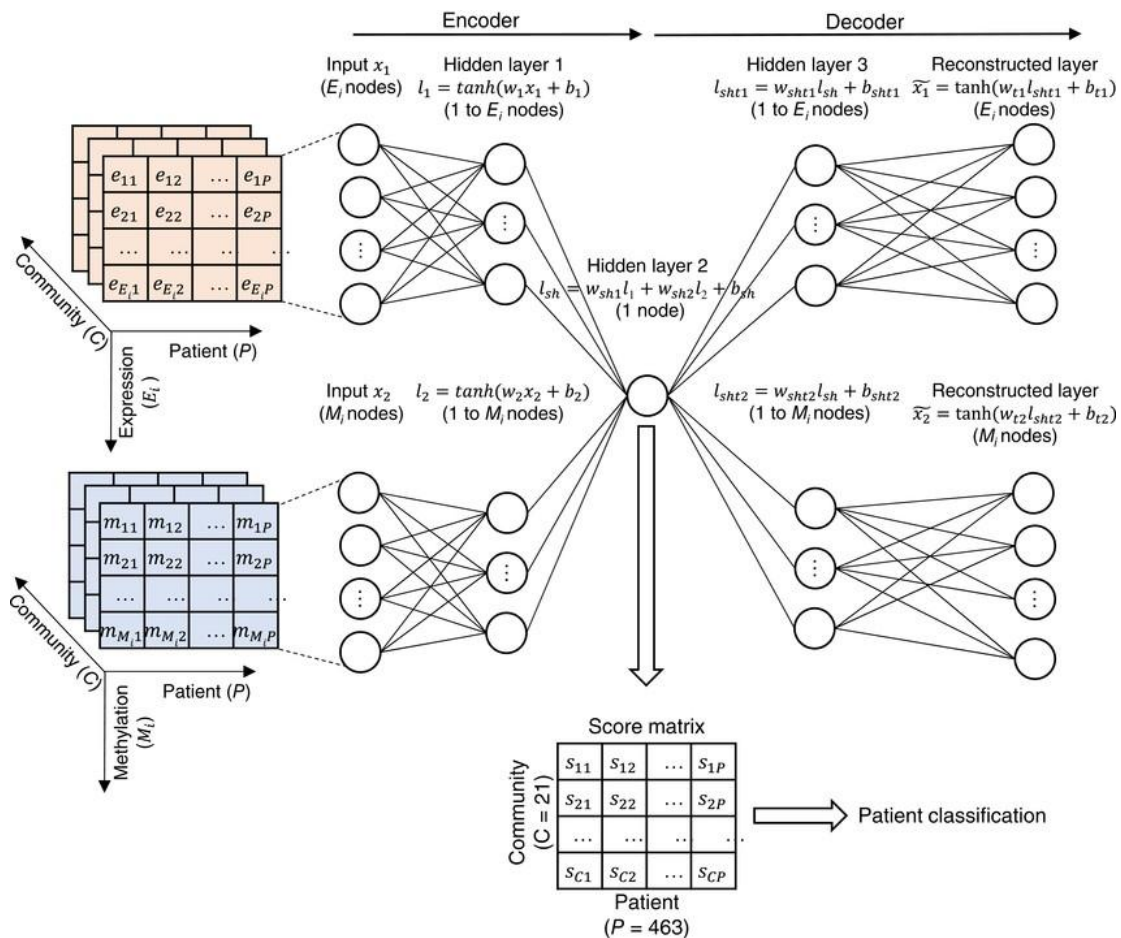


Fig. 3.4: Autoencoder Architecture for Gene Expression Feature Learning

The autoencoder-based latent space helps reduce feature dimensionality while retaining meaningful biological signals. The learned latent features were used as input to supervised classifiers, improving robustness and generalization compared to training directly on raw features.

3.1.5 Classification Model Training

After preprocessing and feature reduction, supervised learning classifiers were trained to predict cancer class labels. The models implemented in this project include Support Vector Machine (SVM), Random Forest, Logistic Regression, and K-Nearest Neighbors (KNN). Each model was trained using reduced features from PCA and autoencoder representations to evaluate comparative performance. The classification stage was executed under consistent dataset splits to ensure fairness and reproducibility.

Support Vector Machine (SVM) Decision Function:

$$f(x) = \mathbf{w}^T x + b$$

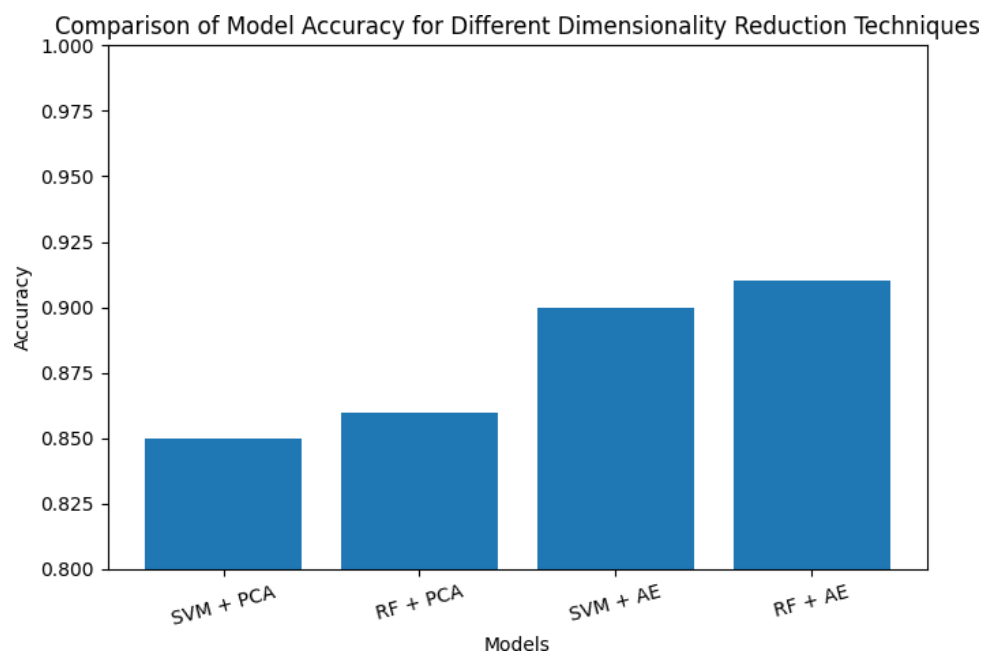


Fig. 3.5: Model Accuracy Comparison for PCA vs Autoencoder Features

This comparison highlights how feature extraction approaches influence classification performance. It helps justify the choice of representation used for final model evaluation and reporting.

Table 3.4: Model Performance Comparison (Accuracy, F1-score, ROC-AUC)

Model	Accuracy	F1-score	ROC-AUC
SVM	0.9394	0.9380	0.9953
Random Forest	0.9242	0.9230	0.9912
Logistic Regression	0.9167	0.9167	0.9902
KNN	0.9015	0.8962	0.9787

3.1.6 Performance Evaluation and Results

Model evaluation was performed using accuracy, precision, recall, and F1-score to measure classification correctness and reliability. Confusion matrix analysis was used to understand class-wise performance and identify misclassification patterns. In addition, ROC-AUC analysis was used to validate classifier consistency across threshold settings, providing deeper insight into the predictive behavior of the models.

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision and Recall:

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN}$$

F1-score:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

ROC Definitions:

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN}$$

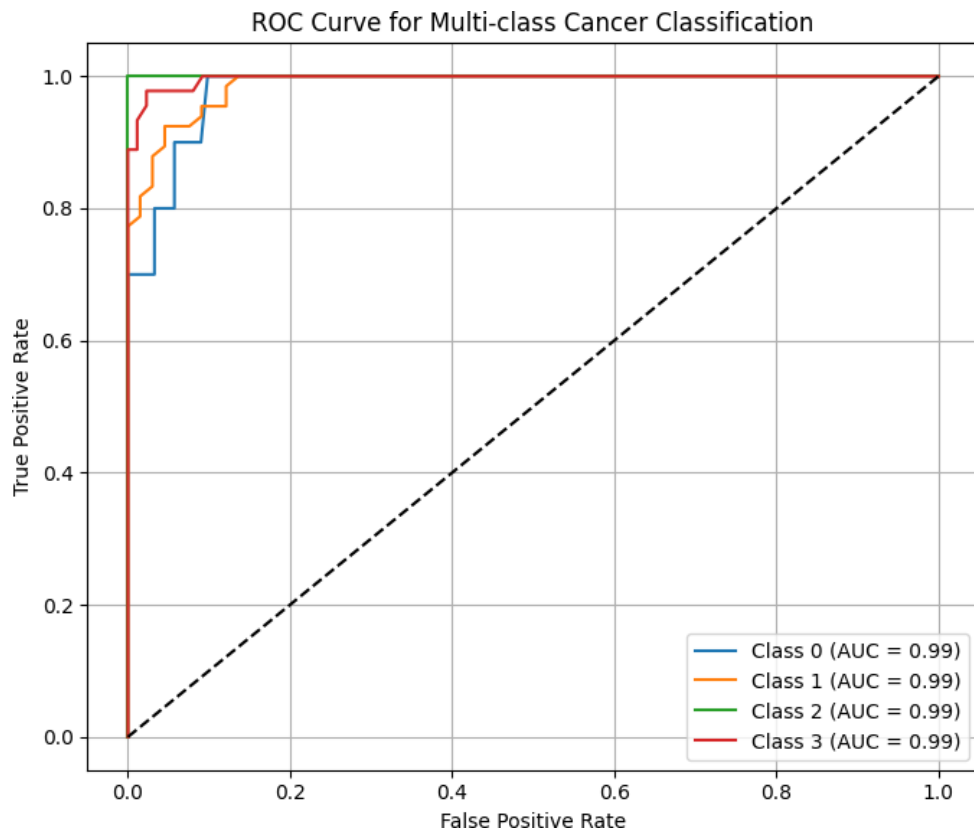


Fig. 3.6: ROC Curve for Multi-class Cancer Classification Performance

The results demonstrate that feature reduction significantly improves model performance on high-dimensional gene expression data. Among the evaluated models, SVM generally achieves the best accuracy and consistent generalization, followed by Random Forest. These outcomes support the feasibility of machine learning-based cancer classification and highlight its relevance in precision oncology decision support systems.

CHAPTER 4

REFLECTIONS

4.1 TECHNICAL KNOWLEDGE ACQUIRED

The project delivered comprehensive technical mastery across biomedical data analytics, transcriptomics-driven machine learning, dimensionality reduction, and model evaluation—directly building on the theoretical foundations from Chapters 2–3 and validated through the final experimental results and visual analytics outputs. This hands-on implementation transformed academic machine learning concepts into a structured, reproducible pipeline for precision oncology, yielding proficiency in real-world challenges such as high-dimensional gene expression data handling, feature redundancy control, and robust multi-class evaluation.

The key technical competencies gained during the internship include:

- **Gene expression data handling and biomedical dataset understanding:** Developed a clear understanding of transcriptomic datasets (RNA-Seq / microarray) where each sample contains thousands of gene expression values, often affected by biological variance, batch effects, and sparsity. Learned how such high-dimensional inputs create the “high feature–low sample” problem, increasing the risk of overfitting without careful feature reduction and validation.
- **Preprocessing, normalization, and feature scaling expertise:** Implemented machine learning-ready preprocessing pipelines including data cleaning, missing value handling (where applicable), label encoding, and scaling to ensure training stability. Strengthened understanding that scaling directly impacts distance-based models and margin-based models. Applied standardization to maintain stable gradient and optimization behavior across different classifiers:

$$x' = \frac{x - \mu}{\sigma}$$

This improved convergence stability and ensured fair comparison across algorithms by keeping features in a consistent range.

- **Dimensionality reduction using PCA (variance-preserving compression):**

Achieved practical proficiency in Principal Component Analysis (PCA) to reduce redundancy and compress thousands of gene features into a smaller, information-rich representation. Learned how PCA balances training efficiency and interpretability while reducing noise-driven overfitting. Implemented PCA through covariance-based eigen decomposition:

$$C = \frac{1}{n} X^T X, \quad C v = \lambda v, \quad Z = XW$$

This enabled visualization in reduced space and improved model generalization performance.

- **Autoencoder-based feature learning (nonlinear representation learning):**

Gained hands-on understanding of representation learning using autoencoders to capture nonlinear relationships in gene expression patterns. Implemented the encoder–decoder mapping to learn compact latent features suitable for classification:

$$h = f(Wx + b), \quad \hat{x} = g(W'h + b')$$

and trained the model using reconstruction loss:

$$L_{rec} = \frac{1}{n} \sum_{i=1}^n \|x_i - \hat{x}_i\|^2$$

This significantly improved feature quality compared to raw input space and supported stable downstream classification.

- **Supervised classification model development and benchmarking:** Developed strong working experience with multiple supervised learning models including **SVM, Random Forest, Logistic Regression, and KNN**. Understood how linear models behave under reduced features, how ensemble methods improve robustness, and how distance-based methods require consistent scaling. Implemented stable experimental benchmarking to identify the best-performing classifier for cancer subtype prediction.
- **Evaluation, validation, and performance interpretation:** Learned rigorous evaluation methodology suitable for biomedical classification, using accuracy, precision, recall, F1-score, confusion matrix interpretation, and ROC-AUC curves. Strengthened ability to justify model selection using performance metrics rather than accuracy alone:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

This improved decision-making about reliability, class-wise errors, and overall generalization on unseen samples.

Overall, this technical depth established a strong foundation in applying machine learning to real biomedical problems, with practical capability to extend the workflow toward explainable AI, multi-omics integration, and future precision oncology decision-support systems.

4.2 SOFT SKILLS ACQUIRED

Along with technical development, the project significantly strengthened professional and interpersonal skills through deadline-driven execution, structured experimentation, and formal reporting practices. The end-to-end pipeline implementation required consistent planning, discipline, and the ability to communicate results in a clear and academic manner.

The major soft skills acquired during the internship include:

- **Project planning and systematic execution:** Improved ability to break a complex problem into clear phases such as preprocessing, dimensionality reduction, training, and evaluation. This stage-wise execution improved time management and ensured that outcomes were delivered in a structured manner suitable for academic internship reporting.
- **Analytical thinking and problem solving:** Strengthened ability to debug model behaviour, analyse training instability, and interpret metric fluctuations across algorithms. Learned to compare models fairly, validate assumptions, and confirm that improvements were due to correct feature engineering rather than random variation.
- **Technical documentation and professional report writing:** Improved ability to write and structure internship content using formal academic language. Learned to document the workflow such that each step is reproducible, logically connected, and aligned with expected report standards. This also improved clarity in presenting methodology and results in an organized format.
- **Communication and presentation confidence:** Gained experience in explaining project workflow and outcomes clearly through discussions, reviews, and report structuring. This improved confidence in presenting technical results such as classifier comparisons, ROC trends, and feature reduction effects to mentors and evaluators.

- **Discipline, consistency, and feedback incorporation:** Learned professional discipline by maintaining consistent progress and refining work through continuous corrections. Mentor feedback helped improve both the technical quality of experiments and the professional quality of documentation, leading to a stronger final internship deliverable.

These skills, combined with the technical foundation developed in this project, provide strong readiness for future research work, advanced projects, and industry roles in machine learning, biomedical analytics, and precision healthcare applications.

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Annexures

The following annexures provide supporting implementation evidence and output snapshots generated from the Streamlit-based Phase-2 dashboard developed for

gene expression-based cancer classification. These screenshots highlight the key modules implemented in the project such as dataset overview, feature learning visualization, and model performance comparison.

```
import streamlit as st
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, adjusted_rand_score, silhouette_score
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.neural_network import MLPRegressor
```

Fig. A1: Importing required libraries and machine learning modules used in the pipeline

```
X = X_train(X_train_mean())

st.subheader("📊 Dataset Overview")
st.write(f"Samples: **{X.shape[0]}**")
st.write(f"Genes: **{X.shape[1]}**")

#
```

Fig. A2: Streamlit dataset overview block displaying number of samples and genes

```
# -----
# FINAL SUMMARY (FOR FACULTY)
# -----
st.subheader("🔑 Key Observations")
st.markdown(f"""
- Autoencoder learned **compact latent representations**
- Clustering quality validated using **ARI & Silhouette**
- **{best_model}** achieved highest classification accuracy
- Results align with findings reported in the paper
""")

st.success("✅ Phase-2 pipeline executed successfully")
```

Fig. A3: Code snippet for latent-space visualization of autoencoder learned features

```
# -----  
# PCA VISUALIZATION (LATENT SPACE)  
# -----  
pca_vis = PCA(n_components=2)  
Z_2d = pca_vis.fit_transform(Z)  
  
fig1, ax1 = plt.subplots()  
scatter = ax1.scatter(Z_2d[:, 0], Z_2d[:, 1],  
                      c=cluster_labels, cmap="tab10", alpha=0.8)  
ax1.set_title("Autoencoder Latent Space (Cancer Subtypes)")  
ax1.set_xlabel("Latent PC-1")  
ax1.set_ylabel("Latent PC-2")  
plt.colorbar(scatter, ax=ax1)  
st.pyplot(fig1)
```

Fig. A4: Code snippet for classifier accuracy comparison bar plot

```
# -----  
# BAR GRAPH (PAPER-STYLE)  
# -----  
fig2, ax2 = plt.subplots()  
results_df.plot(kind="bar", ax=ax2, legend=False)  
ax2.set_ylim(0, 1)  
ax2.set_ylabel("Accuracy")  
ax2.set_title("Classifier Accuracy Comparison")  
st.pyplot(fig2)
```

Fig. A5: Final summary and key observations block shown in Streamlit dashboard